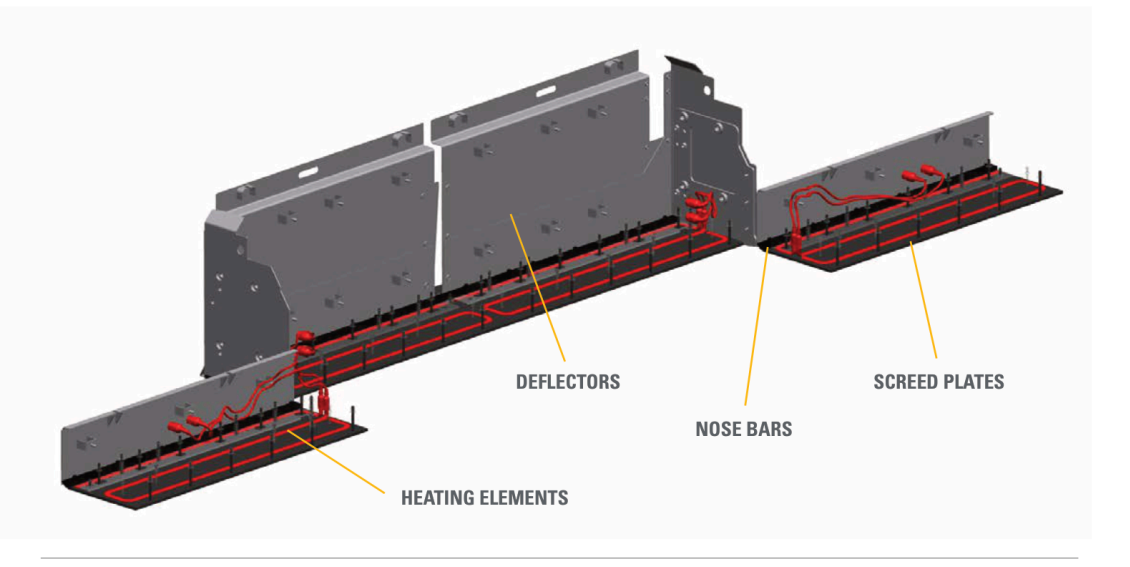
ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Screed control and handwork

Volume 7 · J1, J2, J3, J4, J5, J6, K1, K2, K3

Classroom development edition · 30 October 2026



Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 22). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables.

In this volume

1. How to use the volume and key terms
2. Paving screed and attachments
3. Screed work safety procedures
4. Asphalt screed operator roles and responsibilities
5. Screed maintenance
6. Operate asphalt screed
7. Follow shut down procedures
8. Raker roles and responsibilities
9. Raking safety procedures
10. Place asphalt hot mix
11. Prepare a small-area finishing plan

Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for screed control and handwork. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS

Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT

Audio file / video container / timed web captions; test playback and retain a transcript.

Study chapter · Outline J1

Paving screed and attachments

The screed forms the asphalt layer behind the tractor. Its geometry, material supply, temperature and movement interact. Knowing the parts helps the crew discuss a mat problem without making several unrelated adjustments at once.

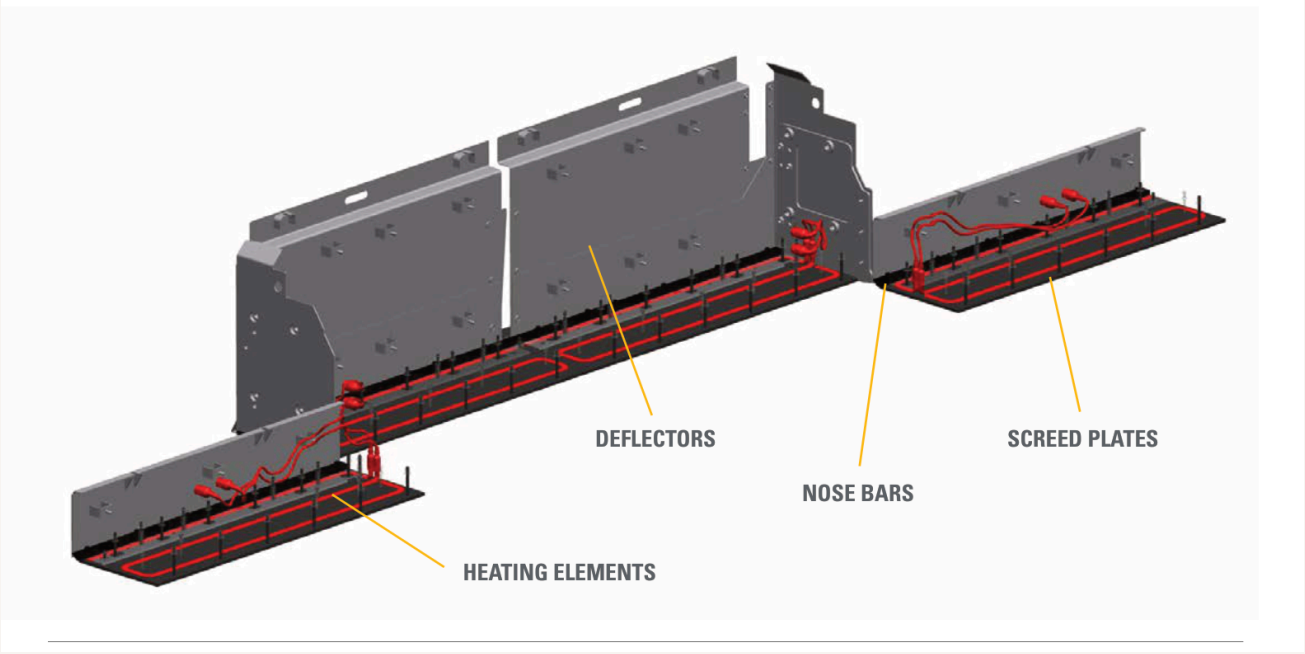
- Screed categories
- Common attachments
- Principal functions

LEARNING STEP 1

Screed categories

A rigid or fixed-width screed uses its assembled width and approved extensions. A variable-width screed changes working width through its designed extension system. Front- and rear-mounted extensions have different arrangements and setup needs. Identify the actual main screed, extensions, plates, heating system, vibration/tamper equipment and control options. Nominal paver size does not establish the fitted screed's full capability.

Screed plates and heating components



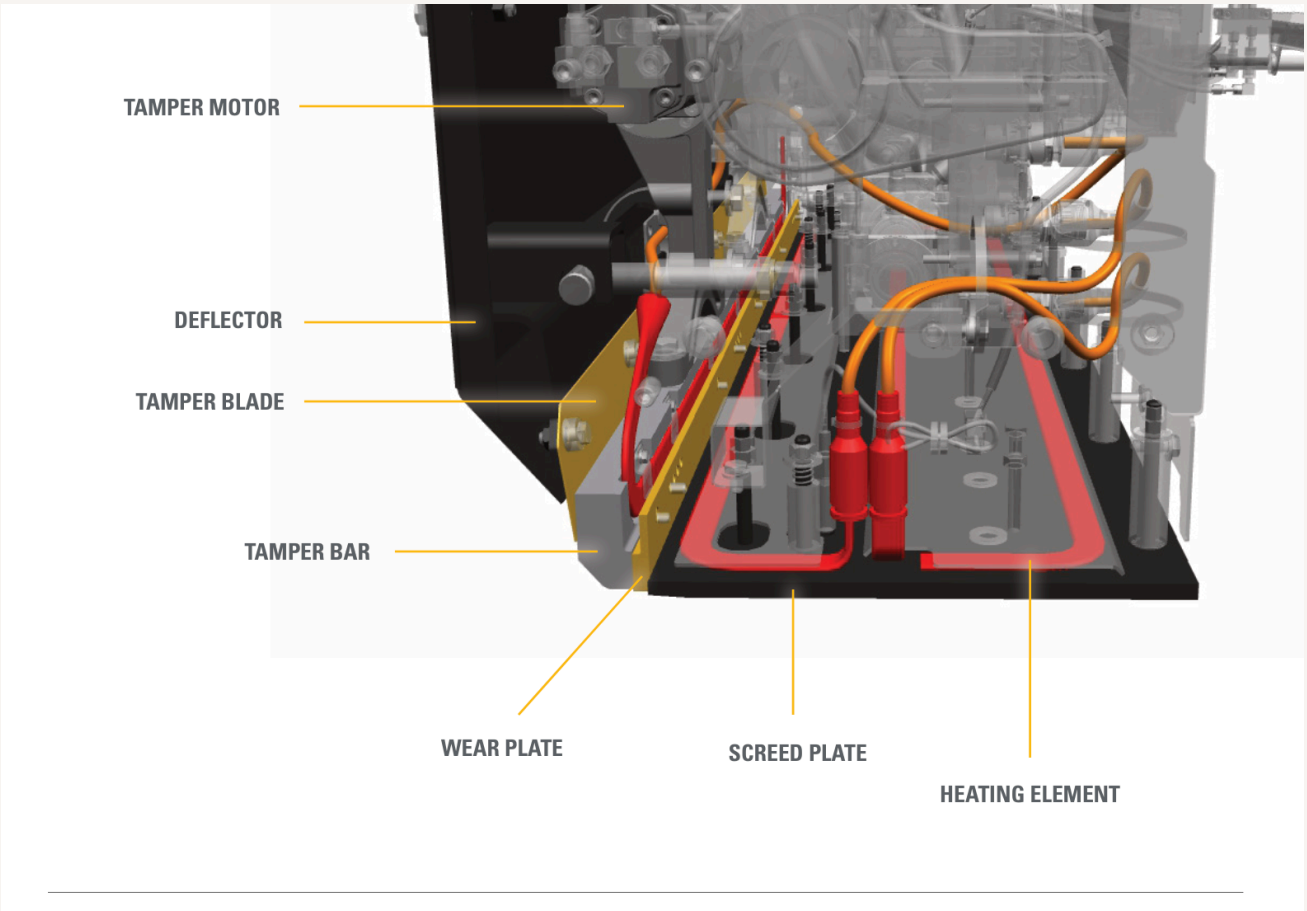
Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 22). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

LEARNING STEP 2

Common attachments

Attachments can include bolt-on extensions, hydraulic extensions, end gates, cutoff shoes, curb-forming equipment and grade/slope controls. A stringline, ski, averaging beam or other reference system supplies different information; a laser or sonic sensor is not an interchangeable reference for every job. Confirm compatibility, mounting, calibration and safe movement envelope. Narrowing a pass with a cutoff attachment affects material distribution and the approved setup.

Look inside a tamper-bar screed



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 28). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

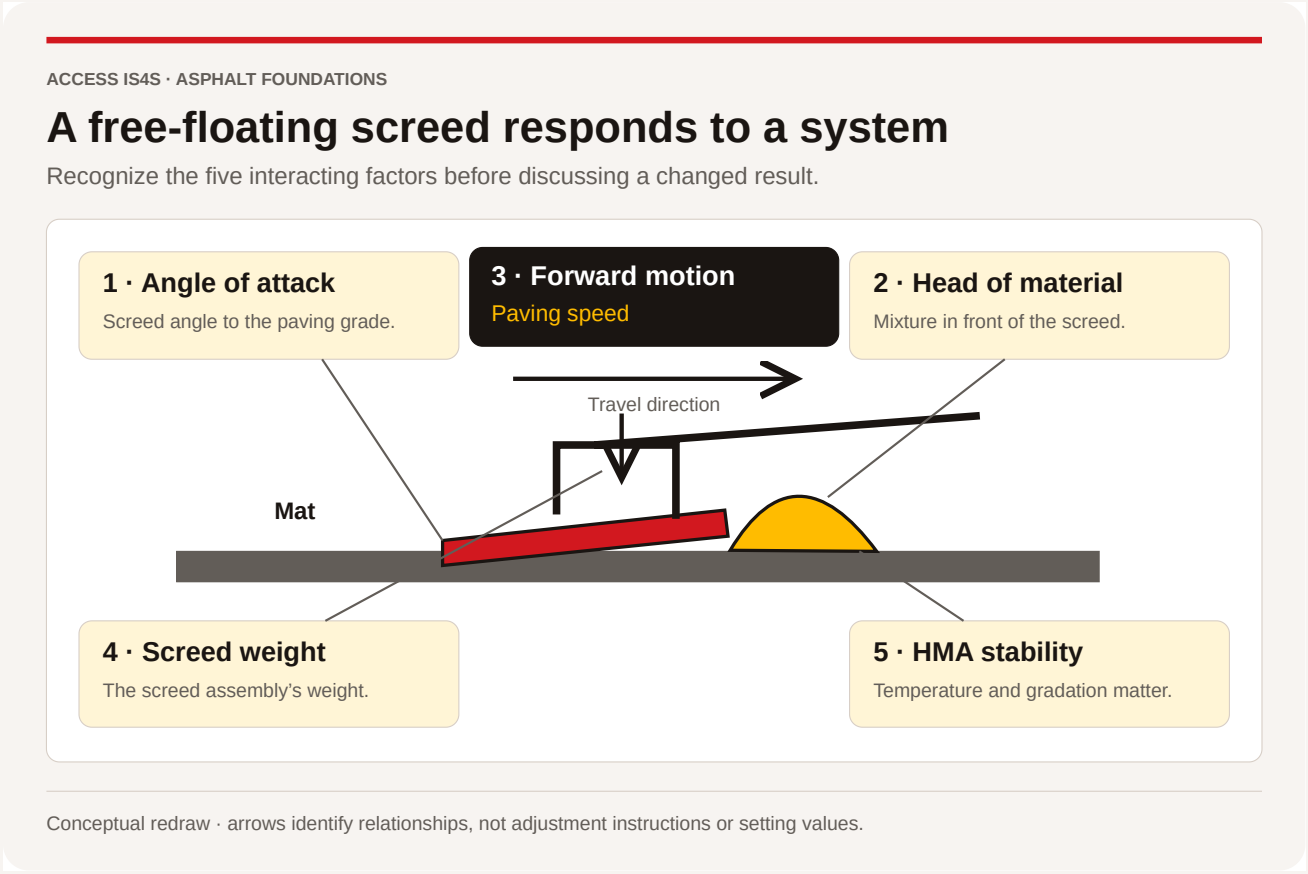
LEARNING STEP 3

Principal functions

The tractor pulls the screed through tow arms; the feed system supplies material in front of it. A floating screed reaches a working balance influenced by tow-point position/angle of attack, material head, speed, mixture and screed condition. A control change takes travel distance to develop its full effect. Do not

repeatedly adjust before observing the response. The main plate and extensions must work together to avoid lines, tearing or uneven thickness. Heat prevents sticking to cold plates; it is not a substitute for correct material temperature or setup.

A free-floating screed responds to a system



Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Original instructional adaptation by ACCESS IS4S (2026). Mirrored and simplified the schematic; grouped the five factors in numbered callouts; omitted adjustment procedures and numeric settings. Source locations: [{"figure": "8.16", "printed_page": 383, "pdf_page": 417}, {"section": "8.4.3, five factors", "printed_page": 384, "pdf_page": 418}]. Original contributors: Volvo Construction Equipment Full source context.

Worked example · An extension line

A visible line follows the inner edge of an extension while the rest of the mat looks consistent.

Think it through.

Record side, station and conditions. Check the relevant extension setup, plate condition, heating and material distribution through the authorized procedure. A line is a clue; do not assume one control or universal adjustment will fix every model.

Try it and explain your thinking

Identify tow arm, main plate, extension, end gate and grade sensor on the supplied figures.

Identify tow arm, main plate, extension, end gate and grade sensor on the supplied figures.

Skills you use: Reading, Communication.

A fictional main width is 2.4 m with 0.6 m extension on each side

A fictional main width is 2.4 m with 0.6 m extension on each side. Find total width.

Skills you use: Numeracy, Problem solving.

Create a checked SVG concept map of the factors affecting a floating screed, using AI if available.

Create a checked SVG concept map of the factors affecting a floating screed, using AI if available.

Skills you use: Digital, Creativity and innovation, Writing.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify screed type and fitted attachments.	0 / 1 / 2	Evidence/help: _____
Explain the material route and floating-screed response.	0 / 1 / 2	Evidence/help: _____
Distinguish an observed mat symptom from a confirmed cause.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, J1, printed p. 82; source PDF pp. 84–84.
- Caterpillar. (Various editions). Original machine component figures, with publication/page/model records. Open publication.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline J2

Screed work safety procedures

Screed work places people close to hot material, moving extensions and the roller path. Safe positions and clear communication must remain effective as the paving width and site conditions change.

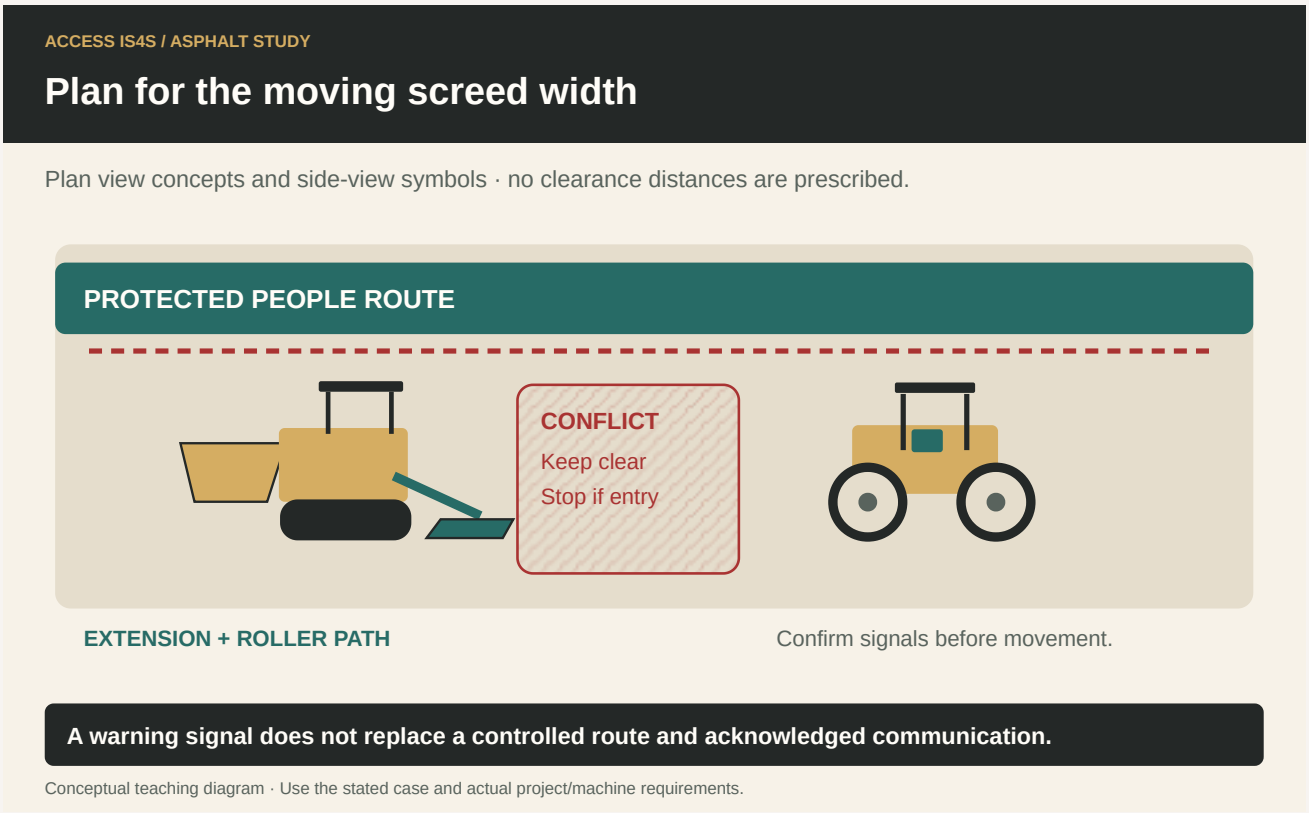
- Interpret job plan
- Machine-specific safety hazards
- Traffic-control requirements
- Proper PPE
- Lockout procedures
- Secure machine for safety

LEARNING STEP 1

Interpret job plan

Review path, pattern, thickness, temperature requirements, crown, joints, obstacles and access. Confirm the current width and any planned changes before extending equipment. Agree on who controls each side, how adjustments are communicated and how the paver/roller crews respond to a stop.

Plan for the moving screed width



Note. ACCESS IS4S. (2026). Plan for the moving screed width [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Machine-specific safety hazards

Identify hot plates and mix, heater energy, fumes, noise, hydraulic pressure, extension pinch zones and moving machinery. Keep feet and clothing clear of the mat and hot surfaces. Wet or unsuitable footwear can worsen burn risk. Plan heat exposure, hydration and breaks through the site program. Do not step into the roller path to inspect the edge; coordinate a safe observation opportunity.

LEARNING STEP 3

Traffic-control requirements

Use the traffic-control plan and assigned trained personnel. The required TCP arrangement depends on the site assessment, not a fixed number in a classroom example. Keep long tools, cables and extension movements inside the controlled work area. Reassess when the work passes an entrance or narrows a public route.

LEARNING STEP 4

Proper PPE

Use the selected footwear, head, eye, hearing, visibility and heat/chemical protection, with training in use and limitations. Check fit and condition, including wet or contaminated clothing. Follow current Part 8 responsibilities and employer arrangements. PPE cannot protect a person who enters an uncontrolled crush zone.

LEARNING STEP 5

Lockout procedures

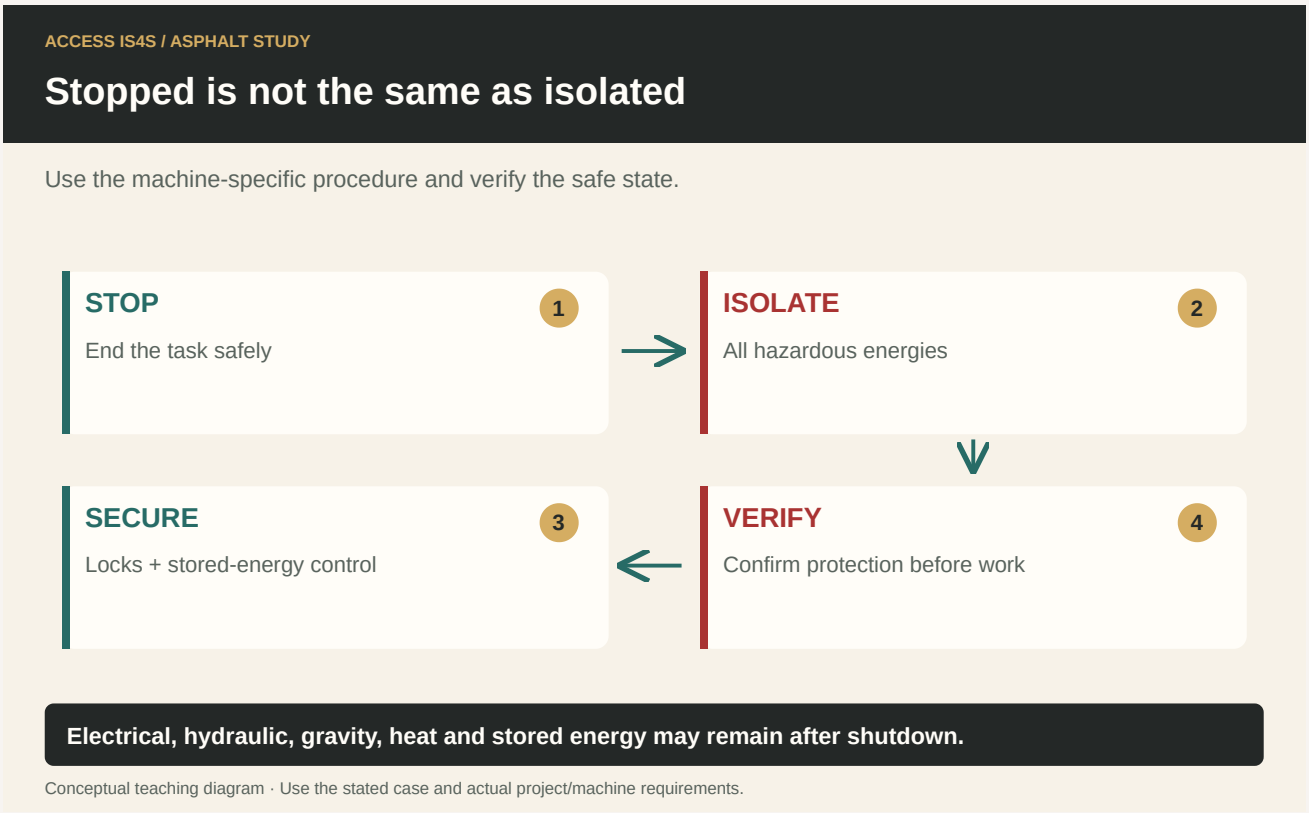
Before service or hazardous access, apply the machine-specific isolation for electrical/heater, fuel where fitted, hydraulic, gravity and stored energy. Support raised components using specified devices and verify the safe state. Screed heating may be electric, propane or another system; identify the actual equipment rather than applying a generic fuel-isolation instruction.

LEARNING STEP 6

Secure machine for safety

At parking or interruption, secure the screed and paver together using the required supports/locks, parking controls and access protection. Prevent unintended extension movement and protect people from residual heat. Report defects and confirm the status with the responsible operator.

Stopped is not the same as isolated



Note. ACCESS IS4S. (2026). Stopped is not the same as isolated [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · Width changes beside a walkway

The plan calls for widening the mat, but the extension would enter a protected walking route.

Think it through.

Hold the extension movement, communicate the conflict and have the responsible lead resolve the work/access arrangement. Brief the change with traffic control and the crew before moving. Do not rely on a shouted warning as the only control.

Try it and explain your thinking

Name three screed hazards that remain after travel stops.

Name three screed hazards that remain after travel stops.

Skills you use: Reading, Problem solving.

A 3.0 m width is widened by 0.45 m on each side

A 3.0 m width is widened by 0.45 m on each side. What new width must the crew plan for?

Skills you use: Numeracy.

Rehearse a stop message when a worker approaches the extension zone.

Rehearse a stop message when a worker approaches the extension zone.

Skills you use: Communication, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Use PPE and safe positions appropriate to heat and movement.	0 / 1 / 2	Evidence/help: _____
Apply actual heater/hydraulic/gravity isolation.	0 / 1 / 2	Evidence/help: _____
Secure screed supports and coordinate paver/traffic roles.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, J2, printed p. 83; source PDF pp. 85–86.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 10: De-energization and Lockout. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 8: Personal Protective Clothing and Equipment. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline J3

Asphalt screed operator roles and responsibilities

The screed operator helps turn the plan into a consistent mat. The role combines observation, measured checks, timely communication and controlled adjustment.

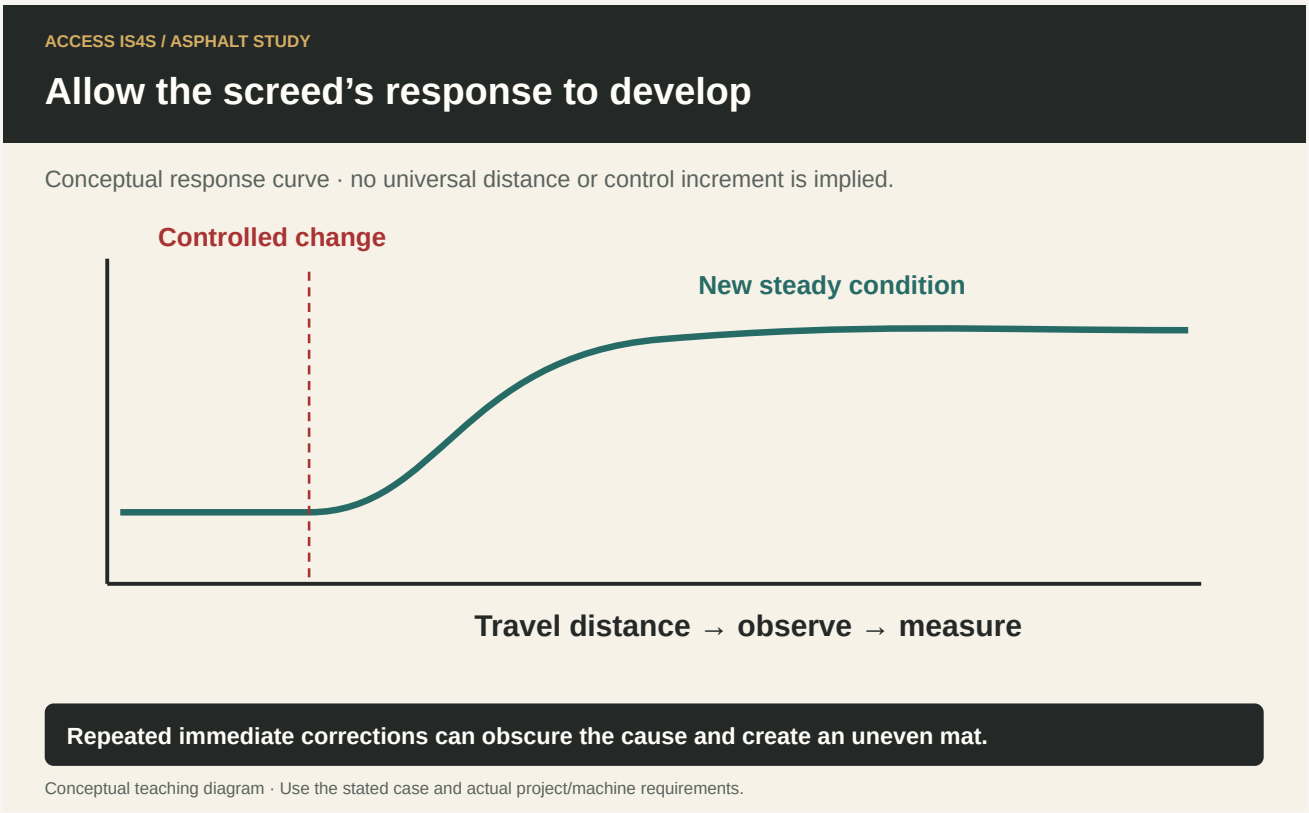
- Plan work procedures
- Maintain mat thickness
- Communication techniques
- Problem solving strategies
- The importance of teamwork
- Importance of adhering to standards

LEARNING STEP 1

Plan work procedures

Read the accepted thickness, width, crown, joint and end-product requirements before setup. Identify where measurements will be made, who records them and who authorizes changes. Confirm the drawing, JMF and quality plan versions. Plan transitions and obstacles before they reach the screed.

Allow the screed’s response to develop



Note. ACCESS IS4S. (2026). Allow the screed’s response to develop [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Maintain mat thickness

Monitor loose mat thickness and width using approved methods and the trial-based relationship to compacted thickness. Material volume depends on area and thickness; delivery and feed must support the intended rate. A thick area and thin area may average to the correct tonnage while both miss the geometry. Keep checks tied to locations.

LEARNING STEP 3

Communication techniques

Communicate with the paver operator and handwork crew using short, specific messages: side, station, observation, requested action and acknowledgement. Share changes in head of material, speed, temperature or width. Do not issue several conflicting adjustments through different people.

LEARNING STEP 4

Problem solving strategies

Investigate a symptom before adjusting. Compare current conditions with the stable setup, identify a plausible cause and make an authorized controlled change. Allow the response to develop over the machine's relevant travel distance, then measure again. Document what changed. Repeated large corrections can produce waves and make the cause harder to identify.

LEARNING STEP 5

The importance of teamwork

The paver operator controls the steady supply/travel process; screed operators control assigned screed functions; rakers manage authorized handwork; quality personnel provide measurements and disposition. These roles depend on one another. Respectful challenge is useful when someone sees a discrepancy: ask for the observation and check it together.

LEARNING STEP 6

Importance of adhering to standards

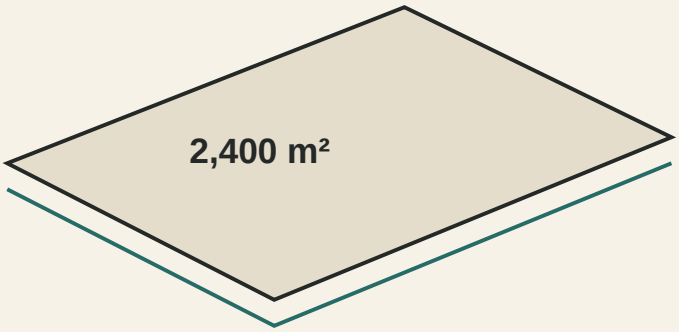
Meet the specified geometry and quantity without treating either excess or shortage as automatically acceptable. Record departures and obtain authorized correction. End-product specifications also include properties beyond screed control, such as density and smoothness. Do not promise full acceptance from a loose-thickness check alone.

Mass divided by area gives application rate

ACCESS IS4S / ASPHALT STUDY

Mass divided by area gives application rate

Fictional lot: 288 t over 2,400 m²; specified rate 120 kg/m².



CALCULATE

288 t = 288,000 kg
288,000 ÷ 2,400
= 120 kg/m²

120 ÷ 120 × 100 = 100% of the specified rate

An average rate does not prove uniform thickness or full acceptance.

Conceptual teaching diagram · Use the stated case and actual project/machine requirements.

Note. ACCESS IS4S. (2026). Mass divided by area gives application rate [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · One controlled change

A mat check is low after a speed change. One person wants more head, another wants a tow-point change and a third wants a different speed.

Think it through.

Pause the conflicting instructions, record the observation and changed condition, and have the responsible lead coordinate a controlled response. Recheck after the relevant response distance. Changing everything at once prevents a useful diagnosis.

Try it and explain your thinking

A 40 m × 3.5 m area at 0.06 m loose thickness has what geometric loose volume

A 40 m × 3.5 m area at 0.06 m loose thickness has what geometric loose volume?

Skills you use: Numeracy, Reading.

Write a message reporting a low right-side loose-depth reading.

Write a message reporting a low right-side loose-depth reading.

Skills you use: Writing, Communication.

Use a shared change log for one setup correction

Use a shared change log for one setup correction. What fields make it useful?

Skills you use: Digital, Collaboration, Problem solving.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

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What the observer checks	Result	Evidence
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Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Plan and monitor geometry with location-based evidence.	0 / 1 / 2	Evidence/help: _____
Coordinate one clear authorized adjustment.	0 / 1 / 2	Evidence/help: _____
Explain the limits of thickness/quantity checks for overall acceptance.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, J3, printed p. 85; source PDF pp. 87–87.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 3: Rights and Responsibilities. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Screed maintenance

Screed maintenance protects plate condition, movement and the quality of the next mat. Use the fitted screed's service instructions and coordinate shared systems with the paver operator.

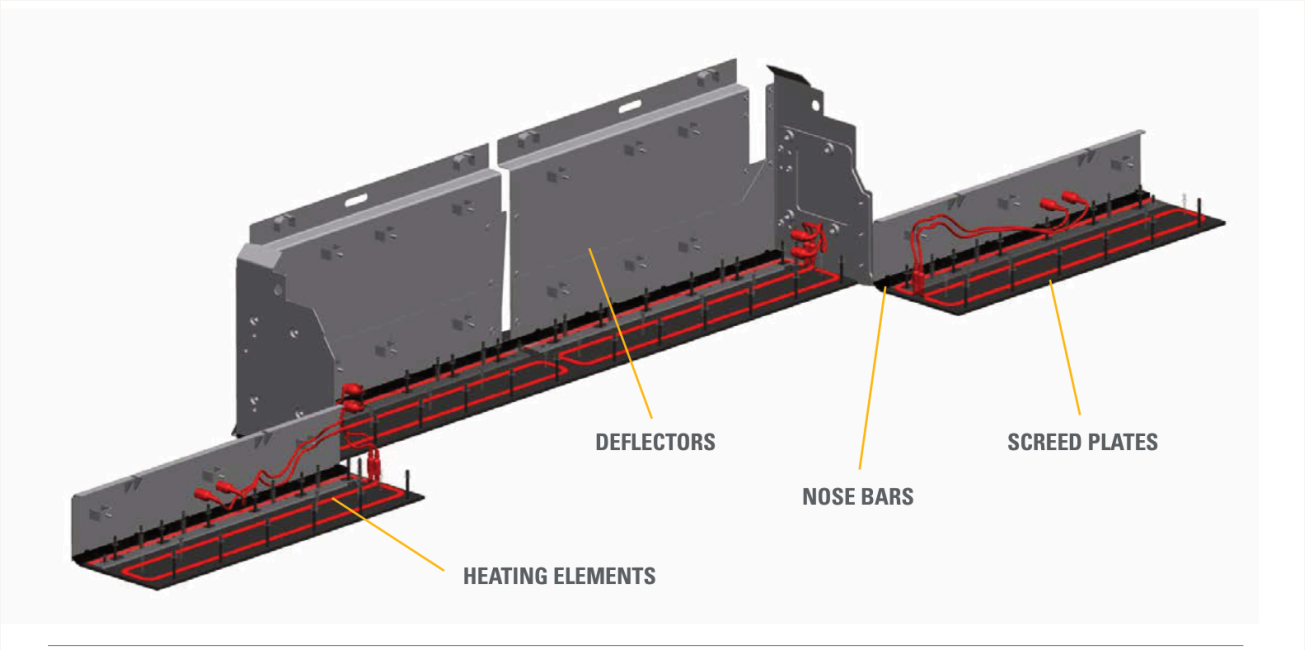
- Service lubrication system (greasing)
- Inspect and clean components as required
- Housekeeping tasks
- Maintain screed plate

LEARNING STEP 1

Service lubrication system (greasing)

Identify lubrication points, including thickness-control screws and extension mechanisms where specified. Confirm lubricant, interval, quantity and safe position. Clean fittings and load the grease gun correctly. Centralized systems and optional fittings change the route; do not assume grease at one point reaches every component. Report damaged fittings and protect the work area from excess lubricant.

Screed plates and heating components



Note. Caterpillar. (2024). Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 22). Faithful crop; original labels retained. Multiple paver/screed models and build numbers; applicability is specified in individual tables. Full source context.

LEARNING STEP 2

Inspect and clean components as required

Inspect accessible components and the paver systems serving the screed, including hydraulic lines, electrical/heating connections and cooling systems where applicable. Radiator, oil-cooler, battery, plug and dipstick checks may belong to the shared tractor maintenance schedule rather than a separate screed engine. Identify what is actually fitted. Use isolation, cooling and approved access; do not open hot pressure systems or diagnose electrical faults beyond authorization.

LEARNING STEP 3

Housekeeping tasks

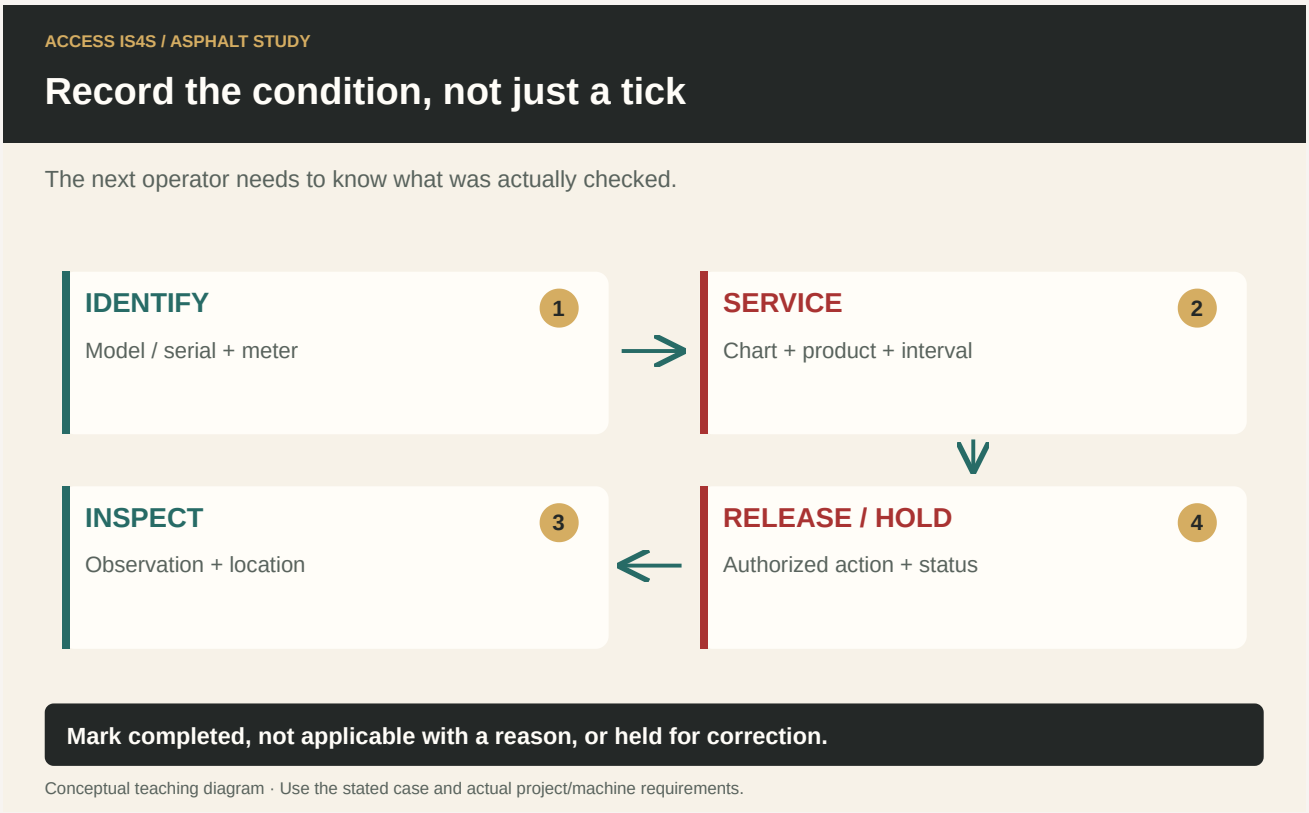
Remove rubbish, buildup and excess grease from safe access surfaces, controls, steps, decks and any fitted cab/windows. Use compatible products and contain waste. Coordinate cleaning with the paver crew so no one restarts or moves a shared system unexpectedly. Restore guards and account for tools.

LEARNING STEP 4

Maintain screed plate

Inspect the screed plate and extension plates for wear, damage, distortion and buildup. Check adjustment and replacement criteria in the actual manual. A scuff or streak in the mat can indicate a plate/setup/material problem; record the pattern and investigate rather than assuming more heat will fix it. Plate replacement and alignment require authorized procedures, support and specified fasteners/settings. Record the condition before and after service.

Record the condition, not just a tick



Note. ACCESS IS4S. (2026). Record the condition, not just a tick [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · A recurring streak

A streak returns along the same line after surface buildup is removed.

Think it through.

Record its position and persistence. Have authorized personnel inspect plate wear/flatness, extension alignment, heating and relevant setup under safe conditions. Do not continue scraping a moving or hot hazard area or guess a replacement tolerance.